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From trigonometry identity,
$(\sin^2 \omega t + \cos^2 \omega t) = 1$

$$E = \frac{1}{2} m\omega^2 A^2 = \text{constant}$$

which gives the law of conservation of total energy. This is depicted in Figure 10.26

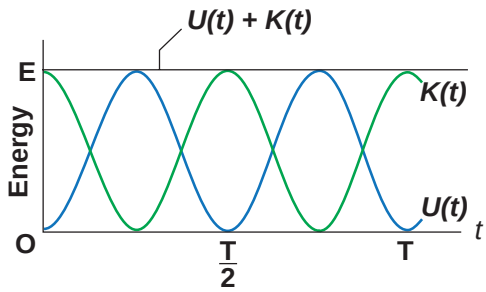


Figure 10.25 Both kinetic energy and potential energy vary but total energy is constant

Thus the amplitude of simple harmonic oscillator, can be expressed in terms of total energy.

$$A = \sqrt{\frac{2E}{m\omega^2}} = \sqrt{\frac{2E}{k}} \qquad (10.75)$$

EXAMPLE 10.15

Write down the kinetic energy and total energy expressions in terms of linear momentum, For one-dimensional case.

Solution

Kinetic energy is $KE = \frac{1}{2} m v_x^2$

Multiply numerator and denominator by m

$$KE = \frac{1}{2m} m^2 v_x^2 = \frac{1}{2m} (m v_x)^2 = \frac{1}{2m} p_x^2$$

where, p_x is the linear momentum of the particle executing simple harmonic motion.

Total energy can be written as sum of kinetic energy and potential energy, therefore, from equation (10.73) and also from equation (10.75), we get

$$E = KE + U(x) = \frac{1}{2m} p_x^2 + \frac{1}{2} m\omega^2 x^2 = \text{constant}$$

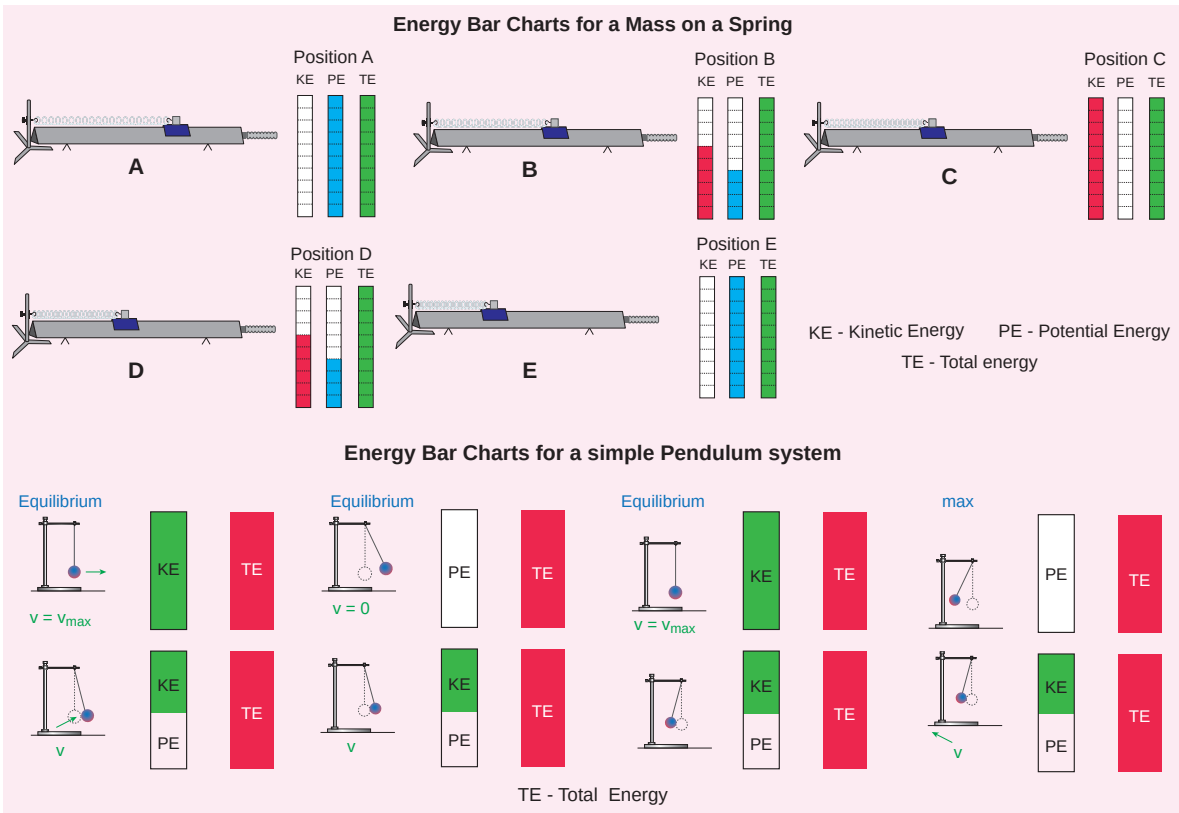


Figure 10.26 Conservation of energy – spring mass system and simple pendulum system